

CONSTRUCTION OS

PRD - Reporting & Analytics

Full Product Requirements Document for Product, UX/UI, Frontend, Backend, Data Engineering, Analytics, QA, Security, DevOps, SRE and Platform Operations Teams

Product purpose: Build the governed, permission-aware analytical layer that converts ConstructionOS operational records into traceable KPIs, dashboards and reports without becoming a second source of truth.

Document control	Value
Document	PRD_Reporting_Analytics
Version	1.0
Status	Implementation-ready functional and technical specification
Audience	Full-stack development, data, analytics, platform, security, QA, SRE and product teams
Primary routes	/analytics, /analytics/dashboards, /analytics/metrics, /reports, /analytics/admin
Core architecture	Operational sources + event/CDC ingestion + analytical warehouse + semantic/KPI layer + permission-aware query and export services
Access model	Deny-by-default, tenant-isolated, company/project scoped, row/field/metric restricted and aggregation-safe
Source-of-truth policy	Operational modules own facts; Reporting owns governed metrics, definitions, snapshots, report definitions and issued results
Integration policy	Self-contained initial release; external BI, data lake and AI integrations deferred
Platform Configuration	Mandatory at domain, dataset, metric, dashboard, widget, report, schedule, export and retention level
Date	2026-08-05

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1. Executive Summary

The Reporting & Analytics Module is the governed intelligence layer for ConstructionOS. It converts operational records from Finance, Projects, Work Progress, Procurement, Inventory, Assets, HR, Payroll, HSE, QA/QC, Contracts, CRM, Units, Documents and Government & Legal into trusted metrics, dashboards, reports, forecasts and issued analytical records.

The module must not duplicate operational state, run uncontrolled formulas, or expose restricted information through aggregation. Every KPI must be explainable, versioned and traceable to its source records. Every dashboard and report must recalculate under the viewer's permissions, while issued reports preserve their original metric versions, filters and result snapshots.

Permanent rule: Operational modules own facts. Reporting defines trusted metrics. Dashboards present them. Every result must remain explainable and traceable to its original records.

1.1 Required outcomes

- One governed KPI catalogue shared by dashboards, embedded widgets, reports and exports.
- Permission-safe analytics across corporate groups, companies, projects, departments and Joint Ventures.
- Reproducible historical reporting through snapshots, metric versions, currency-rate versions and issued report records.
- Fast executive drill-down from a group KPI to company, project, metric components and individual source records.
- Self-service dashboard and report creation without arbitrary code execution or bypassing the semantic layer.
- Visible data quality, refresh status and source lineage on every critical metric.
- Asynchronous, auditable PDF, Excel and CSV exports for large datasets.

1.2 Success indicators

Outcome	Target behavior
Trust	Every KPI shows definition, formula, source, period, version and refresh state.
Security	No user can infer restricted data from a dashboard, export, average or small-group aggregation.
Traceability	Every analytical result drills to governed components and authorized source records.
Reproducibility	An issued historical report can be regenerated using preserved metric and rate versions.
Performance	Common dashboards render quickly without running heavy queries against operational databases.
Resilience	A failed widget or data source does not make the entire dashboard unavailable.

2. Product Vision, Outcomes and Failure Risks

2.1 Product vision

ConstructionOS must provide one analytical language across every company and project. Executives, project managers, finance teams, site teams and specialist functions should read the same governed metrics while receiving only the detail they are authorized to access.

2.2 Business outcomes

- Earlier detection of cost, schedule, procurement, quality, safety and legal risks.
- Comparable project and company performance using consistent metric definitions.
- Faster month-end, project review and executive reporting.
- Reduced spreadsheet dependency and contradictory management reports.
- Clear accountability for metric ownership, data quality and report publication.

- Better planning through transparent forecasts and scenario comparison.

2.3 Failure risks the design must prevent

Risk	Required prevention
Multiple definitions for the same KPI	KPI Catalogue with active versions, owners and effective dates.
Dashboard totals differ from Finance	Source reconciliation and Finance-owned accounting facts.
Restricted salary or legal data inferred	Aggregation privacy, minimum group sizes and metric restrictions.
Past reports change after corrections	Issued result snapshots and preserved metric versions.
Operational databases overloaded	Separate analytical storage, pre-aggregations and asynchronous exports.
Attractive charts without explanation	Mandatory drill-down, formula, lineage and data-quality metadata.
AI or forecasts treated as fact	Method, assumptions, confidence and actual-vs-forecast history shown.

3. Scope, Non-Goals and Assumptions

3.1 In scope

- Analytical ingestion, reporting warehouse, semantic layer, KPI Catalogue and permission projections.
- Executive, company, project, finance, procurement, inventory, asset, workforce, HSE, quality, legal, contract and CRM analytics.
- Dashboard Builder, Report Builder, Report Library, saved filters, scheduled reports and exports.
- Data quality, lineage, refresh status, snapshots, currency translation and consolidation.
- Real-time, near-real-time and scheduled analytical refresh modes.
- Responsive desktop, tablet and mobile viewing.
- APIs, background jobs, observability, audit, security and testing.

3.2 Non-goals

- Replacing operational modules or writing business records from analytics.
- Replacing statutory accounting, payroll, project scheduling or contract administration.
- Allowing unrestricted SQL, arbitrary scripts or ungoverned formulas in the first release.
- Using analytics automatically as individual employee performance judgments.
- Requiring external BI platforms, data lakes or AI services for initial operation.
- Implementing final Approvals & Workflow orchestration.

3.3 Assumptions

- Operational modules expose stable IDs, tenant/company scope, status history and domain events.
- Company, project, department and user permissions are available through central authorization services.
- Finance provides authoritative postings, periods and currency context.
- Platform Configuration is available for feature, metric and report enablement.

4. Permanent Architectural Principles

ID	Principle	Required behavior
RA-P01	Operational ownership	Reporting reads and analyses; it never mutates source business records.
RA-P02	Explainability	Every KPI provides formula, source fields, filters, version and drill-down.

ID	Principle	Required behavior
RA-P03	Permission before aggregation	Restricted rows and fields are excluded before totals are calculated.
RA-P04	Reproducibility	Snapshots, rates, metric versions and report definitions are effective-dated and preserved.
RA-P05	Semantic consistency	Dashboards, embedded widgets and exports use the same metric definitions.
RA-P06	Failure isolation	One failed widget, dataset or source does not fail the entire dashboard.
RA-P07	Data quality visibility	Stale, incomplete or unreconciled data is never presented as fully trusted.
RA-P08	Self-contained core	Core analytics works without external BI or AI providers.

5. Domain Ownership and Source-of-Truth Boundaries

Domain	Authoritative owner	Reporting responsibility
Accounting, invoices and payments	Finance	Aggregate, compare, consolidate and drill to Finance records.
Project identity and hierarchy	Projects	Use project dimensions and authorized hierarchy history.
Physical completion and quantities	Work Progress	Calculate progress, productivity and forecast metrics.
Contracts and obligations	Contracts	Analyse value, milestones, exposure and performance.
Purchasing workflow	Procurement	Analyse spend, lead time, competition, commitment and supplier performance.
Stock quantity and movement	Inventory	Analyse availability, valuation, accuracy and turnover.
Asset lifecycle	Assets	Analyse utilisation, cost, downtime and compliance.
Employment and payroll	HR & Payroll	Analyse workforce and payroll under strict confidentiality.
Safety records	HSE	Analyse leading and lagging safety indicators.
Quality acceptance and handover	QA/QC	Analyse inspections, NCRs, defects and readiness.
Regulatory and legal records	Government & Legal	Analyse permits, cases, obligations and exposure.
Files and issued reports	Documents	Store issued report files and preserve controlled revisions.
Metric definitions and report results	Reporting & Analytics	Own KPI Catalogue, dashboards, report definitions, snapshots and issued results.

6. Personas, Roles and Access Model

Persona	Primary needs	Special controls
Executive	Group/company/project health, trends and exceptions	Consolidated access only where authorized.
Project Manager	Progress, schedule, cost, procurement, risk and handover	Project-scoped drill-down.
Finance	Financial statements, budgets, forecasts and reconciliation	Account, company and period restrictions.
Commercial/Contracts	Contract value, claims, obligations and profitability	Confidential commercial scope.
Procurement	Spend, supplier, lead time and delivery analytics	Supplier and project scope.
Warehouse/Assets	Stock and equipment operational analytics	Location and custody scope.
HR/Payroll	Headcount, payroll, attendance and labour allocation	Field-level salary and medical restrictions.
HSE/QA	Safety, quality, corrective action and readiness	Site/project specialist access.
Legal/Compliance	Permit, case, fine and obligation analytics	Matter-level confidentiality.
Report Author	Build governed reports and dashboards	Dataset and publishing capabilities.
Data Steward	Own metric and data quality definitions	No automatic business-record edit rights.
Platform Admin	Configure datasets, metrics and jobs	Does not automatically receive business data access.

Access rule: Sharing a dashboard or report never grants access to its data. The viewer receives a permission-filtered execution or an explicitly authorized issued result.

7. Information Architecture

- Executive Overview
- My Dashboards
- Company Analytics
- Project Analytics
- Financial Analytics
- Commercial Analytics
- Procurement Analytics
- Inventory Analytics
- Asset Analytics
- Workforce Analytics
- HSE Analytics
- QA/QC Analytics
- Legal & Compliance Analytics
- Contracts Analytics
- Client & Sales Analytics
- Report Library
- Custom Report Builder
- Dashboard Builder
- KPI Catalogue
- Scheduled Reports
- Data Quality
- Analytics Administration
- Settings

7.1 Embedded analytics

Governed analytics widgets may appear inside Company, Project, Unit, Client, Contract, Procurement Package, Warehouse, Asset, Employee, HSE Site, QA/QC Handover and Legal Case pages. Embedded widgets must use the same semantic metrics and permission logic as the central analytics pages.

7.2 Primary routes

```
/analytics
/analytics/executive
/analytics/projects/{projectId}
/analytics/companies/{companyId}
/analytics/dashboards/{dashboardId}
/analytics/metrics/{metricId}
/reports/{reportId}
/reports/executions/{executionId}
/analytics/admin
```

8. Reference Data Architecture

Operational Databases

- > Transactional Outboxes / Domain Events / CDC
- > Analytical Ingestion & Validation
- > Staging / Raw History
- > Transformations & Reconciliation
- > Reporting Warehouse / Materialized Aggregates
- > Semantic Layer & KPI Catalogue
- > Permission-Aware Query Service
- > Dashboards / Reports / Exports / Embedded Widgets

8.1 Architectural requirements

ID	Area	Requirement	Priority	Acceptance signal
RA-001	Ingestion	Operational transactions and analytical events must be decoupled through durable ingestion.	Must	Source outages do not corrupt published reports.
RA-002	Warehouse	Complex analytics must execute against analytical storage, not operational databases.	Must	Operational latency remains stable during report load.
RA-003	Semantic	All official widgets and reports must resolve metrics through governed definitions.	Must	Same KPI returns consistent values across surfaces.
RA-004	Permissions	Query plans must apply permission projections before aggregation.	Must	Restricted records cannot influence visible totals.
RA-005	Rebuild	Analytical datasets must support replay, backfill and deterministic rebuild.	Must	A clean rebuild reconciles to operational sources.

9. Executive Overview

9.1 Core KPI groups

Group	Example KPIs
Financial	Revenue, gross profit, net profit, cash, receivables, payables, forecast cash requirement.
Projects	Active projects, delayed projects, over-budget projects, forecast completion and backlog.
Commercial	Contracted value, pipeline, sales, reservations, certified value and claims exposure.
Operations	Procurement commitment, inventory value, asset availability and workforce capacity.
Control	HSE exposure, quality defects, legal restrictions, overdue obligations and handover readiness.

9.2 Attention centre

- Project cost variance above threshold
- Critical schedule delay
- Negative cash projection
- Contract obligation overdue
- Permit affecting active work
- Critical HSE incident
- Handover blocked
- Payroll exception
- Inventory shortage affecting progress
- Supplier failure affecting delivery
- Asset downtime affecting critical work

9.3 Drill path

```

Group KPI
  -> Company
    -> Project / Department / Business Unit
      -> Metric component
        -> Aggregated record set
          -> Authorized source record

```

10. KPI Catalogue and Governance

10.1 KPI definition fields

- KPI ID and name
- Business description
- Metric owner
- Source modules and datasets
- Formula and aggregation method
- Dimensions and supported filters
- Included and excluded statuses
- Currency and unit treatment
- Refresh frequency
- Targets and thresholds
- Metric version and effective dates
- Data-quality requirements
- Confidentiality class
- Drill-down route
- Lifecycle status

10.2 KPI lifecycle

Draft -> Under Review -> Active -> Revised -> Deprecated -> Archived

10.3 Versioning rules

- A formula, source-field, status-filter or currency-treatment change creates a new KPI version.
- Historical report executions retain the KPI version used at execution time.
- Only one default active version may apply for a given effective date and scope.
- Deprecated metrics remain readable for historical reports but cannot be newly selected unless explicitly allowed.

10.4 Example KPI specification

Field	Example value
Name	Project Cost Variance Percentage
Definition	Difference between actual/committed cost and approved budget divided by approved budget.
Formula	$(\text{Actual} + \text{Committed} - \text{Approved Budget}) / \text{Approved Budget} \times 100$
Sources	Finance postings, Procurement commitments, approved project budget.
Dimensions	Company, project, cost code, work package, period.
Exclusions	Voided postings, cancelled purchase orders, superseded budgets.
Drill-down	Project -> cost code -> source posting/commitment.
Quality gate	Budget and Finance period must reconcile.

11. Metric Categories and Definitions

Category	Required metric families
Financial	Revenue, cost, margin, cash flow, budget variance, forecast variance, receivable/payable days, profitability.
Project controls	Planned/actual/accepted/certified progress, schedule variance, cost variance, forecast completion, constraints.
Procurement	Commitment, category spend, lead time, late delivery, RFQ response, price variance, savings, exceptions.
Inventory	Value, available/reserved stock, shortages, turnover, stock accuracy, dead and slow-moving stock.
Assets	Utilisation, availability, downtime, maintenance cost, inspection compliance and operating cost.
Workforce	Headcount, labour hours, overtime, absence, payroll cost, allocation and competence compliance.
HSE	Incidents, near misses, inspections, permits, corrective actions, training and stop-work duration.
QA/QC	Pass rate, NCR, defect recurrence, reinspection, snag closure, handover and warranty defects.
Legal	Permits, expiries, overdue obligations, cases, fines, legal cost and restrictions.
Commercial	Leads, conversion, pipeline value, unit availability, sales, reservations, contracts and collections.

12. Dashboard Architecture

12.1 Dashboard types

- Personal
- Executive
- Group
- Company
- Project
- Department
- Team
- Site
- Operational
- Client-facing
- Custom

12.2 Widget types

- KPI card
- Trend line
- Bar/column/area chart
- Waterfall
- Progress ring/gauge
- Heat map
- Geographic map
- Gantt summary
- Cost and cash-flow curves
- Table
- Pivot table
- Ranked list
- Risk matrix
- Timeline
- Data-quality indicator
- Text/explanation
- Source-record list

12.3 Widget contract

Field	Requirement
Metric reference	Official metric ID and version selection policy.
Filters	Dashboard, widget and viewer filters with explicit precedence.
Comparison	Previous period, baseline, budget, target or forecast.
Permission behavior	Hide, redact or recompute according to metric policy.
Refresh	Display last refresh and stale state.
Drill-down	Open metric explanation and authorized source records.
Failure state	Independent error with retry; other widgets continue loading.

13. Dashboard Builder

- Add, resize, reorder and remove widgets.
- Select governed metrics and datasets.
- Configure filters and comparison periods.
- Set targets and thresholds where allowed.

- Save personal views and duplicate dashboards.
- Share dashboard definitions without granting data.
- Publish controlled company/project templates.
- Export and schedule supported dashboard layouts.

13.1 Ownership states

Type	Behavior
Personal	Editable by owner; viewer permissions always apply.
Shared company/project	Published by authorized report author; definition versioned.
Platform template	Configured centrally and instantiated by scope.
Locked executive	Layout controlled; users may change only allowed filters.

13.2 Functional requirements

ID	Area	Requirement	Priority	Acceptance signal
RA-101	Builder	Users must be able to preview a widget before adding it.	Must	Preview returns governed sample under current permissions.
RA-102	Builder	Widget layout changes must be saved with optimistic version checks.	Must	Concurrent edits return a conflict instead of overwriting.
RA-103	Sharing	Shared dashboard definitions must not contain cached restricted data.	Must	Viewer receives a fresh permission-filtered execution.
RA-104	Templates	Administrators must publish reusable dashboard templates by business category.	Should	New companies/projects can instantiate a configured template.

14. Project Analytics

-
- Project summary
 - Progress and schedule
 - Cost and cash flow
 - Contracts
 - Procurement
 - Materials
 - Assets
 - Labour
 - Quality
 - HSE
 - Legal readiness
 - Handover
 - Forecast

Comparison	Required inputs
Baseline vs current schedule	Approved baseline, current schedule, status date and critical path.
Planned vs actual progress	Baseline curve, measured quantities, accepted progress and status date.
Budget vs committed/actual	Approved budget, procurement commitment and Finance postings.
Certified vs paid	Progress certifications, client invoices and receipts.

Comparison	Required inputs
Labour planned vs used	Workforce plan, timesheets and Work Progress usage.
Materials planned vs consumed	BOQ/material plan, Inventory issues and Work Progress consumption.

Project health score: A configurable combined score may exist, but every component must remain visible and a serious safety, legal or cash issue must never be hidden by averaging.

15. Financial and Consolidation Analytics

- Profit and loss
- Balance sheet
- Cash flow
- Budget performance
- Forecast
- Receivables
- Payables
- Banking
- Project/unit/contract profitability
- Department cost
- Company and group consolidation

15.1 Currency treatment

- Transaction currency
- Company base currency
- Group reporting currency
- Exchange-rate date
- Rate source
- Historical rate
- Closing rate
- Average rate
- Stored rate version

15.2 Consolidation rules

- Company ownership percentage
- Intercompany matching and elimination
- Joint Venture presentation
- Minority-interest readiness
- Consolidated budget and cash flow
- Company comparison

16. Commercial, CRM and Contract Analytics

Area	Required analytics
CRM	Leads, source, response time, conversion, opportunity pipeline, win/loss and client activity.
Units/Sales	Availability, reservation, sale value, price per area, collections and cancellations.
Contracts	Contract value, amendments, obligations, milestones, claims, guarantees, certification and payment.

Area	Required analytics
Clients	Lifetime value, purchases, payments, support activity and relationship history.

17. Procurement Analytics

- Purchase request cycle time
- RFQ competition and response rate
- Quotation comparison
- Purchase commitment
- Spend by category/supplier/project
- Price variance
- Lead time
- Late delivery rate
- Procurement savings
- Three-way-match exceptions
- Supplier quality and performance

17.1 Package analytics

Construction procurement packages must support consolidated budget, commitment, award, delivery and contract analysis across multiple purchase requests, suppliers and purchase orders.

18. Inventory Analytics

- Inventory value
- Available/reserved/in-transit stock
- Low and out-of-stock exposure
- Overstock
- Dead and slow-moving stock
- Stock accuracy
- Count variance
- Turnover
- Expiry
- Warehouse comparison
- Project consumption
- Shortage impact on work progress

18.1 Valuation visibility

Reports must distinguish operational quantity from Finance-owned accounting postings and identify the valuation method and period used.

19. Asset Analytics

- Asset register value
- Availability
- Utilisation
- Downtime
- Maintenance cost
- Cost per operating hour
- Inspection compliance
- Fuel/energy consumption
- Warranty and insurance exposure
- Project allocation

- Replacement forecast

20. Workforce, HR and Payroll Analytics

- Headcount
- Vacancy and recruitment funnel
- New hires and turnover
- Attendance and absence
- Overtime
- Timesheet completion
- Training and certification compliance
- Workforce allocation
- Payroll cost
- Project labour cost
- Benefits
- Contract expiry

Privacy safeguard: Workforce analytics must not automatically become individual performance judgments. Salary, medical, disciplinary and small-group metrics require separate restrictions and aggregation thresholds.

21. Work Progress and Schedule Analytics

- Planned vs actual quantities
- Accepted/certified/paid progress
- Production rates
- Lookahead reliability
- Constraint age
- Delay causes
- Recovery performance
- Crew productivity
- Material and equipment usage
- Forecast completion
- Evidence completeness

21.1 Progress states

Submitted, accepted, certified and paid progress must remain separate measures. Dashboards must not treat unverified or uncertified quantities as contractual completion.

22. HSE Analytics

Indicator type	Examples
Leading	Hazards reported, near misses, inspections, permits, training, toolbox talks, corrective action closure.
Lagging	Injuries, lost-time incidents, environmental events, regulatory breaches and work-stoppage duration.
Readiness	Worker eligibility, asset compliance, active permits, critical hazards and emergency readiness.

23. QA/QC and Handover Analytics

- Inspection pass rate
- ITP completion
- NCR rate and age
- Corrective action closure
- Defect recurrence
- Snag and punch closure
- Reinspection rate
- Commissioning status
- Handover readiness
- Document completeness
- Warranty defects

24. Government, Legal and Compliance Analytics

- Permit register and expiry
- Applications awaiting response
- Regulatory inspections
- Overdue obligations
- Notices and orders
- Fines and penalties
- Active cases and stages
- Appeal deadlines
- Insurance gaps
- Legal cost
- Work-blocking legal restrictions

25. Custom Report Builder

25.1 Workflow

1. Select governed subject/dataset.
2. Select fields and relationships.
3. Add approved calculated fields.
4. Configure filters.
5. Configure grouping and sorting.
6. Add totals and comparisons.
7. Select table or visualization.
8. Preview under current permissions.
9. Save, publish, export or schedule.

25.2 Supported subjects

- Projects
- Units
- Clients
- Contracts
- Finance
- Procurement
- Suppliers
- Inventory
- Assets
- Employees
- Payroll
- Tasks
- Work Progress

- QA/QC
- HSE
- Legal
- Documents
- Meetings
- Configured governed datasets

25.3 Calculated field safeguards

- Approved arithmetic and statistical functions only.
- No arbitrary SQL, JavaScript, Python or external calls.
- Type validation and division-by-zero handling.
- Formula versioning and dependency tracking.
- Confidential inputs cannot be exposed through derived fields.

26. Report Library and Lifecycle

- Executive
- Financial
- Project
- Commercial
- Procurement
- Inventory
- Assets
- HR
- Payroll
- HSE
- QA/QC
- Legal
- Audit
- Client
- Regulatory
- Custom

26.1 Report definition fields

- Report ID
- Name and description
- Owner
- Dataset
- Filters
- Layout
- Columns
- Grouping
- Metric references
- Version
- Access scope
- Export settings
- Schedule
- Retention
- Activity history

26.2 Lifecycle

Draft -> Testing -> Published -> Revised -> Deprecated -> Archived

27. Scheduled Reports

Capability	Requirement
Recurrence	Daily, weekly, monthly, quarterly, yearly, fiscal close and custom recurrence.
Destination	In-app report inbox and Documents folder in core release.
Recipients	Users and roles already authorized for the selected data.
Execution context	Permissions rechecked at execution time.
Failure	Retry policy, failure log and notification-ready event.
Outputs	PDF, Excel or CSV according to report definition.

28. Report Outputs and Issued Reports

28.1 Output formats

- PDF
- Excel
- CSV
- PNG for charts
- Structured JSON for approved internal/API use
- Print layout

28.2 Export controls

- Watermark
- Generated-by identity
- Generation date
- Filters and period
- KPI version
- Confidentiality label
- Page numbering
- Source references
- Export audit
- Expiration where supported

28.3 Live vs issued

Type	Behavior
Live report	Recalculates under the current viewer permissions and current dataset refresh.
Issued report	Frozen result with issuance date, preserved filters, metric versions, rate versions and controlled document revision.

29. Drill-Down, Traceability and Data Lineage

```

Dashboard
-> Widget
  -> KPI definition and version
    -> Metric components
      -> Aggregated record set
        -> Authorized source record
  
```

29.1 Lineage metadata

- Metric version
- Transformation version
- Source module/entity/field
- Source event
- Ingestion job
- Calculation time
- Data owner
- Currency rate version
- Snapshot ID

29.2 Drill-down requirements

ID	Area	Requirement	Priority	Acceptance signal
RA-201	Traceability	Every critical KPI must expose its formula and source modules.	Must	KPI details drawer is available.
RA-202	Traceability	Users must drill to authorized source records without duplicating them.	Must	Source opens in owning module.
RA-203	Lineage	Issued reports must preserve lineage metadata.	Must	Historical report identifies metric and transformation versions.

30. Analytical Warehouse and Semantic Model

30.1 Storage layers

- Raw immutable event/CDC history
- Validated staging
- Conformed dimensions
- Fact tables
- Snapshot facts
- Materialized aggregates
- Semantic metric views
- Query cache

30.2 Semantic layer responsibilities

- Business-friendly fields
- Metric formulas
- Join rules
- Default status filters
- Currency and unit behavior
- Time intelligence
- Permission predicates
- Lineage metadata
- Dataset certification

31. Fact, Dimension and Snapshot Models

31.1 Example fact tables

FactFinanceTransaction
FactProjectProgress

```
FactProcurementSpend
FactInventoryMovement
FactAssetUtilisation
FactLabourHour
FactPayroll
FactHSEIncident
FactQualityInspection
FactLegalObligation
FactContractObligation
FactUnitSale
```

31.2 Example dimensions

```
DimDate
DimCompany
DimProject
DimDepartment
DimUser
DimSupplier
DimClient
DimContract
DimAsset
DimProduct
DimLocation
DimCurrency
DimStatus
DimWorkPackage
```

31.3 Snapshot types

- Daily
- Weekly
- Month-end
- Fiscal-period
- Project baseline
- Contract baseline
- Controlled user-defined snapshot

32. Data Ingestion and Refresh

32.1 Ingestion modes

- Domain-event stream
- Transactional outbox relay
- Change-data capture where required
- Scheduled extraction for legacy sources
- Controlled manual import for migration only

32.2 Refresh modes

- Real-time event update
- Near real-time
- Scheduled batch
- Period close
- Administrative rebuild

32.3 Refresh metadata

- Last source event
- Last ingestion time

- Last transformation time
- Last successful refresh
- Refresh status
- Data delay
- Incomplete-source warning

32.4 Late events and backfill

The pipeline must accept late-arriving source events, recalculate affected partitions and preserve issued snapshots. Late corrections may update live analytics but must not silently replace an issued report.

33. Data Quality and Reconciliation

33.1 Quality dimensions

- Completeness
- Accuracy
- Consistency
- Timeliness
- Uniqueness
- Validity
- Reconciliation

33.2 Example rules

- Progress quantity exceeds contract quantity
- Financial totals do not reconcile
- Project missing baseline
- Asset missing ownership
- Negative stock
- Payroll result missing Finance export
- Contract payment exceeds contract value
- Permit state inconsistent with expiry
- Orphaned dimension

33.3 Quality states

- Trusted
- Acceptable
- Warning
- Incomplete
- Failed
- Stale
- Unknown

Visibility rule: A data-quality warning must appear directly on the affected KPI or report, not only in an administration page.

34. Permissions and Aggregation Privacy

34.1 Security layers

1. Tenant isolation
2. Corporate group scope
3. Company scope
4. Project scope
5. Department scope
6. Record access

- 7. Field restrictions
- 8. Metric confidentiality
- 9. Export permission

34.2 Aggregation privacy

- Minimum group-size rules for configured workforce metrics.
- Suppression of restricted categories and totals that enable subtraction attacks.
- No salary average or legal exposure when the visible cohort is below threshold.
- Viewer-specific recalculation instead of filtering a precomputed unauthorized total.
- Confidential metric labels may be hidden even when a safe aggregate is visible.

34.3 Permission change behavior

- Cached query results invalidated.
- Saved dashboards remain but restricted widgets recalculate or become unavailable.
- Exports require authorization at generation time.
- Issued report access remains controlled by its document/report permission.

35. Multi-Company Consolidation and Currency Translation

35.1 Consolidation levels

Project -> Department -> Legal Company -> Corporate Group

35.2 Required capabilities

- Multi-company reporting
- Intercompany identification and elimination
- Currency translation
- Ownership percentage
- Joint Venture reporting
- Minority-interest readiness
- Consolidated budget
- Consolidated cash flow
- Company comparison

35.3 Reproducibility

Consolidated results must store the company hierarchy, ownership percentages, elimination rules and currency-rate snapshot used for the reporting period.

36. Forecasting and Benchmarking

36.1 Initial forecasting methods

- Linear trend
- Remaining-duration forecast
- Budget burn forecast
- Cash-flow forecast
- Procurement lead-time forecast
- Labour-demand forecast

36.2 Forecast explanation

- Input data
- Method
- Assumptions
- Forecast date

- Confidence indicator
- Manual overrides
- Actual vs forecast history

36.3 Benchmarking

- Project vs project
- Company vs company
- Department vs department
- Supplier vs supplier
- Asset vs asset
- Current vs previous period
- Actual vs budget/baseline/forecast/target

AI policy: AI-generated summaries and predictive models are future roadmap items. Transparent deterministic methods are required for the core release.

37. Documents and Notifications Integration

37.1 Documents

- Issued reports stored through Documents.
- Document Passport, revision, watermark, retention, required reading and legal hold apply.
- One issued report may appear in multiple folders through references without duplication.

37.2 Suggested folder structure

```
Reports/
  Executive/
  Companies/
  Projects/
  Finance/
  Procurement/
  Inventory/
  Assets/
  HR and Payroll/
  HSE/
  QA-QC/
  Government and Legal/
  Client Reports/
  Scheduled Reports/
```

37.3 Notifications

- KPI threshold exceeded
- Report generated
- Report failed
- Scheduled report ready
- Data-quality failure
- Dashboard stale
- Forecast variance exceeded

Notifications owns delivery. Reporting publishes notification-ready events and must not implement channel logic.

38. Data Model

AnalyticsDataset

DatasetField
 DatasetRelationship
 MetricDefinition
 MetricVersion
 MetricThreshold
 DimensionDefinition
 Dashboard
 DashboardWidget
 DashboardFilter
 DashboardShare
 ReportDefinition
 ReportVersion
 ReportExecution
 ReportResult
 ReportSnapshot
 ReportSchedule
 ReportRecipient
 ExportJob
 DataSource
 DataIngestionJob
 TransformationJob
 DataQualityRule
 DataQualityResult
 DataLineageRecord
 CurrencyRateSnapshot
 ConsolidationRule
 ForecastDefinition
 ForecastResult
 AnalyticsPermissionProjection
 AnalyticsActivity
 AnalyticsMetricSnapshot

38.1 Common entity fields

- tenant_id
- company/project scope
- stable business ID
- status
- record version
- created/updated metadata
- effective dates where applicable
- confidentiality
- archive state
- audit correlation ID

39. Backend Service Architecture

-
- Analytics Ingestion Service
 - Transformation Service
 - Reporting Warehouse
 - Semantic Layer Service
 - KPI Catalogue Service
 - Analytics Query Service
 - Dashboard Service
 - Report Definition Service
 - Report Execution Service
 - Snapshot Service
 - Consolidation Service
 - Currency Translation Service
 - Data Quality Service
 - Data Lineage Service

- Forecast Service
- Scheduled Report Service
- Export Service
- Reporting Search Service
- Analytics Audit Service
- Cache Service

39.1 Processing principles

- Idempotent ingestion
- Incremental transformations
- Deterministic rebuild
- Late-event handling
- Backfill jobs
- Versioned transformations
- Correlation IDs
- Retry and dead-letter handling
- Partition isolation

40. Domain Events and Integration Contracts

40.1 Events consumed

```
finance.journal.posted
finance.payment.recorded
projects.project.updated
work_progress.measurement.accepted
procurement.purchase_order.issued
inventory.movement.posted
assets.assignment.changed
hr.payroll.run.locked
hse.incident.reported
qaqc.inspection.completed
legal.permit.status_changed
contracts.contract.amended
crm.opportunity.stage_changed
```

40.2 Events published

```
analytics.dataset.refreshed
analytics.metric.threshold_exceeded
analytics.data_quality.failed
reports.execution.completed
reports.execution.failed
reports.issued
reports.schedule.failed
analytics.snapshot.finalized
```

40.3 Contract rules

- Explicit schema version
- Tenant and company scope
- Source aggregate version
- Idempotency key
- Occurred and published timestamps
- Correlation and causation IDs
- Sensitivity class

41. REST API Contract

41.1 Dashboards

```
GET    /analytics/dashboards
POST   /analytics/dashboards
GET    /analytics/dashboards/{id}
PATCH /analytics/dashboards/{id}
POST   /analytics/dashboards/{id}/duplicate
POST   /analytics/dashboards/{id}/share
POST   /analytics/dashboards/{id}/publish
```

41.2 Metrics

```
GET    /analytics/metrics
POST   /analytics/metrics
GET    /analytics/metrics/{id}
POST   /analytics/metrics/{id}/versions
POST   /analytics/metrics/{id}/activate
GET    /analytics/metrics/{id}/lineage
POST   /analytics/metrics/{id}/preview
```

41.3 Reports

```
GET    /reports
POST   /reports
GET    /reports/{id}
PATCH /reports/{id}
POST   /reports/{id}/execute
POST   /reports/{id}/publish
POST   /reports/{id}/issue
POST   /reports/{id}/export
GET    /report-executions/{id}
```

41.4 Scheduling and data quality

```
POST   /reports/{id}/schedules
PATCH /report-schedules/{id}
DELETE /report-schedules/{id}
GET    /report-schedules/{id}/runs
GET    /analytics/data-quality
POST   /analytics/data-quality/rules
GET    /analytics/data-quality/issues
POST   /analytics/data-quality/issues/{id}/recheck
```

41.5 API requirements

- Cursor pagination
- Idempotency for write and execution requests
- Optimistic version control
- Asynchronous job references
- Permission context
- Filter validation
- Export size limits
- Rate limiting
- Audit correlation ID

42. Frontend Architecture and State

```
ExecutiveAnalyticsPage
DashboardGrid
DashboardWidget
KPIWidget
ChartWidget
TableWidget
TimelineWidget
RiskMatrixWidget
DashboardBuilder
WidgetLibrary
MetricSelector
FilterBuilder
ComparisonSelector
ReportLibrary
ReportBuilder
DatasetSelector
FieldSelector
FormulaBuilder
ReportPreview
ReportScheduleEditor
ExportPanel
KPIDetailDrawer
MetricLineageViewer
SourceRecordDrawer
DataQualityPanel
AnalyticsAdminPage
```

42.1 State rules

- URL-based filters
- Saved personal state
- Server-state cache
- Cancelable queries
- Independent widget loading
- Permission-invalidated cache
- Virtualized large tables
- Asynchronous export progress
- No optimistic issued report creation

43. Search, Filters and Saved Views

43.1 Common filters

- Date/fiscal period
- Company
- Project
- Department
- Team
- Client
- Supplier
- Contract
- Status
- Currency
- Location
- Owner
- Risk/quality state
- Custom configured dimensions

43.2 Filter precedence

Platform default -> Dashboard filter -> Widget filter -> Viewer temporary filter

43.3 Saved views

Saved views store filter and layout preferences only. They do not store permission-bypassing result sets. A saved view must re-execute under the current user and current access scope.

44. Responsive and Mobile Behavior

Viewport	Required behavior
Desktop	Multi-widget grids, persistent filters, report builder, side-by-side comparison and drill-down drawers.
Tablet	Two-column dashboard, collapsible filters, adaptive charts and full-screen report preview.
Mobile	KPI-first single-column view, compact tables, filter sheet, saved dashboards and export requests.

Complex report authoring may be restricted to desktop/tablet, but report viewing, filtering, drill-down and export request status must remain mobile accessible.

45. Premium UX and Accessibility

- Smooth independent widget loading
- Subtle KPI count transitions
- Interactive drill-down
- Comparison period transitions
- Expandable metric explanations
- Lineage drawer
- Data-quality badges
- Export progress
- Dashboard layout animation
- Reduced-motion mode

45.1 Accessibility

- Keyboard navigation
- Visible focus states
- Chart alternatives and data tables
- No color-only status communication
- Screen-reader labels
- High-contrast readiness
- Accessible PDF templates
- Locale-safe number/date formatting

46. Platform Configuration

- Enabled analytics domains
- KPI catalogue
- Metric formulas
- Thresholds and targets
- Reporting periods/fiscal calendars
- Currency rules

- Consolidation rules
- Dashboard/report templates
- Dataset availability
- Export formats
- Scheduling limits
- Retention
- Snapshot frequency
- Data-quality rules
- Confidentiality
- Minimum aggregation size
- Forecast methods
- Custom fields
- Notification hooks

Configuration boundary: Platform Configuration enables and shapes capabilities. Authorization separately determines who can see or use them.

47. Security, Privacy and Confidentiality

- Tenant and company isolation
- Project/department scope
- Row-level security
- Field masking
- Metric confidentiality
- Aggregation privacy
- Export permission
- Encryption in transit and at rest
- Secure query logging
- No production data in development
- Signed/expiring download links
- Sensitive view audit

47.1 Restricted domains

- Executive compensation
- Payroll and bank details
- Medical/disciplinary data
- Legal privileged matters
- Restricted financial accounts
- Confidential clients/contracts
- Security/HSE investigations

48. Audit, Observability and Operations

48.1 Audit events

- KPI created/changed/activated
- Dashboard published
- Report shared/executed/issued
- Export generated/downloaded
- Schedule created
- Snapshot finalized
- Data-quality rule changed
- Consolidation rule changed
- Permission denied
- Sensitive report viewed

48.2 Technical observability

- Ingestion lag
- Transformation duration
- Query latency
- Cache hit rate
- Widget failure rate
- Export queue time
- Schedule success rate
- Warehouse storage
- Dataset freshness
- Dead-letter count
- Rebuild duration

49. Performance, Scalability and Availability

Area	Target
Cached executive dashboard	First meaningful content under 3 seconds under normal load.
Common filters	Response under 2 seconds for pre-aggregated datasets.
Drill-down tables	Initial page under 3 seconds with pagination/virtualization.
Large exports	Asynchronous; user receives job state and completion event.
Widget isolation	One failed widget does not fail the page.
Availability	Read analytics remain available during isolated source-module outages when data is cached and freshness is shown.

49.1 Techniques

- Pre-aggregations
- Materialized views
- Incremental refresh
- Query cache
- Partitioning
- Columnar storage
- Pagination
- Virtualization
- Background export
- Query limits
- Workload isolation

50. Validation, Error Handling and Concurrency

Condition	Required response
Invalid metric formula	Reject with field-level errors and no new active version.
Conflicting dashboard edit	Return version conflict and preserve both users' changes for review.
Stale dataset	Render visible stale indicator; configurable hard block for controlled reports.
Partial source failure	Load unaffected widgets and identify unavailable data source.
Export too large	Queue async export or require narrower filters.
Permission changed during execution	Abort or re-evaluate before result delivery.

Condition	Required response
Snapshot finalization conflict	Allow only one successful finalization with idempotency.

51. Retention, Deletion, Snapshots and Legal Hold

- Live query caches expire according to policy.
- Report definitions and metric versions follow configured retention.
- Issued reports and snapshots are immutable except controlled superseding records.
- Legal hold prevents deletion of relevant report results, exports and lineage.
- Source record deletion/archive propagates according to source retention and analytical anonymization rules.
- Personal dashboard deletion does not delete source data or shared metric definitions.

52. Testing Strategy and Critical Scenarios

52.1 Required test suites

- Metric formula tests
- Aggregation tests
- Currency conversion
- Consolidation and elimination
- Historical reproducibility
- Snapshot immutability
- Permission filtering
- Aggregation privacy
- Data-quality rules
- Drill-down traceability
- Scheduled reports
- Export accuracy
- Large-data performance
- Late-event handling
- Backfill
- Multi-company/JV isolation
- Mobile responsiveness
- Accessibility
- Failure recovery

52.2 Critical scenarios

- KPI definition changes after an issued report.
- User loses project access after saving a dashboard.
- Company changes reporting currency.
- Late Finance posting arrives after month-end snapshot.
- Project hierarchy changes mid-year.
- Intercompany transaction exists in only one company.
- Source record is corrected through reversal.
- One dashboard widget fails while others remain available.
- Scheduled report includes a restricted employee record.
- Export job runs after initiating user is deactivated.
- Data-quality issue is resolved after a report was issued.
- Small-group workforce filter would expose one person's salary.

53. Acceptance Criteria

ID	Area	Acceptance criterion	Evidence
AC-001	Catalogue	KPI Catalogue supports lifecycle, versions, owners, effective dates and lineage.	Pass when verified by automated or documented QA evidence.
AC-002	Consistency	The same metric returns the same result across dashboard, report and embedded widget for the same context.	Pass when verified by automated or documented QA evidence.
AC-003	Permissions	Restricted rows are removed before aggregation.	Pass when verified by automated or documented QA evidence.
AC-004	Privacy	Configured minimum aggregation size suppresses sensitive small cohorts.	Pass when verified by automated or documented QA evidence.
AC-005	Drill-down	Critical KPIs drill to authorized source records.	Pass when verified by automated or documented QA evidence.
AC-006	Quality	Every KPI exposes refresh and data-quality state.	Pass when verified by automated or documented QA evidence.
AC-007	Snapshots	Finalized snapshots are immutable and reproducible.	Pass when verified by automated or documented QA evidence.
AC-008	Issued reports	Issued reports preserve filters, metric versions, rate versions and result data.	Pass when verified by automated or documented QA evidence.
AC-009	Dashboards	Widgets load independently and failure is isolated.	Pass when verified by automated or documented QA evidence.
AC-010	Builder	Dashboard Builder supports governed metrics, filters, layout and preview.	Pass when verified by automated or documented QA evidence.
AC-011	Reports	Report Builder supports governed datasets, fields, formulas, grouping and preview.	Pass when verified by automated or documented QA evidence.
AC-012	Scheduling	Scheduled reports execute under refreshed permissions.	Pass when verified by automated or documented QA evidence.

ID	Area	Acceptance criterion	Evidence
AC-013	Exports	PDF/Excel/CSV exports are asynchronous when large and fully audited.	Pass when verified by automated or documented QA evidence.
AC-014	Finance	Financial analytics reconcile to authoritative Finance totals.	Pass when verified by automated or documented QA evidence.
AC-015	Consolidation	Intercompany elimination is visible and reproducible.	Pass when verified by automated or documented QA evidence.
AC-016	Currency	Consolidated reports preserve exchange-rate versions.	Pass when verified by automated or documented QA evidence.
AC-017	Project	Project analytics separate planned, actual, accepted, certified and paid progress.	Pass when verified by automated or documented QA evidence.
AC-018	Forecast	Forecasts expose inputs, method, assumptions and confidence.	Pass when verified by automated or documented QA evidence.
AC-019	Lineage	Metric lineage reaches source module/entity/field.	Pass when verified by automated or documented QA evidence.
AC-020	Data quality	Failed quality rules appear on affected dashboards.	Pass when verified by automated or documented QA evidence.
AC-021	Security	Shared dashboard definitions never grant data access.	Pass when verified by automated or documented QA evidence.
AC-022	Mobile	Users can view, filter and drill into dashboards on mobile.	Pass when verified by automated or documented QA evidence.
AC-023	Performance	Cached executive dashboards meet performance target under normal load.	Pass when verified by automated or documented QA evidence.
AC-024	Resilience	Source outage shows stale status without corrupting historical reports.	Pass when verified by automated or documented QA evidence.
AC-025	Audit	Sensitive views, report issuance and exports are auditable.	Pass when verified by automated or documented QA evidence.
AC-026	Configuration	Every domain, metric, dashboard, widget, report and export honors Platform Configuration.	Pass when verified by automated or documented QA evidence.

54. Delivery Phases

Phase	Scope
Phase 1 - Governed foundation	Reporting warehouse, ingestion, KPI Catalogue, core dashboards, project/financial analytics, Report Library, exports, drill-down, permissions and audit.
Phase 2 - Self-service	Dashboard Builder, Custom Report Builder, saved filters, schedules, Data Quality, lineage and historical snapshots.
Phase 3 - Enterprise intelligence	Consolidation, currency translation, intercompany elimination, forecasting, benchmarking and advanced executive analytics.
Future roadmap	External BI/data-lake connectors, AI summaries, natural-language analytics, predictive analytics and anomaly detection.

55. Migration and Adoption

55.1 Migration inputs

- Existing spreadsheet KPI definitions
- Legacy reports and filters
- Company/project hierarchies
- Historical financial and progress snapshots
- Currency rates
- User dashboard preferences where practical

55.2 Migration controls

- Definition mapping and owner sign-off
- Reconciliation against legacy reports
- Historical cut-off dates
- Duplicate metric detection
- Permission mapping
- Data-quality baseline
- Parallel reporting period before decommissioning spreadsheets

55.3 Adoption

- Role-based dashboard templates
- Metric glossary
- Report-author training
- Data steward ownership
- Executive review cadence
- Feedback and deprecated-report process

56. Page and Flow Registry

Page/flow	Primary purpose
Executive Overview	Cross-company and project health, attention centre and drill-down.
Dashboard Page	View and interact with configured widgets.
Dashboard Builder	Create and publish governed dashboard definitions.
KPI Catalogue	Define, version, activate and explain official metrics.
KPI Detail	Formula, thresholds, quality, lineage and source drill-down.
Report Library	Discover published reports by domain and scope.
Report Builder	Create governed table/chart reports.
Report Execution	Run, filter, inspect and export a report.
Scheduled Reports	Configure and review recurring executions.
Data Quality	Review failed rules, affected datasets and refresh status.
Analytics Admin	Manage datasets, jobs, transformations and access projections.

57. Risks and Open Decisions

Topic	Decision required
Analytical datastore	Select initial warehouse technology and scaling model.
CDC coverage	Define which modules require CDC in addition to domain events.
Refresh SLAs	Confirm real-time vs near-real-time expectations by metric domain.
Aggregation thresholds	Define default privacy minimums by sensitive domain and jurisdiction.
Currency rates	Select authoritative rate source and period rules.
Metric ownership	Assign product/business owners for each KPI family.
Issued report retention	Define retention by report class and legal requirements.
Forecast governance	Define which forecast methods may be published as official.

58. Definition of Done

- All must-priority functional requirements are implemented and tested.
- KPI Catalogue and semantic definitions are versioned and documented.
- Core operational modules reconcile to analytical results.
- Permission and aggregation-privacy tests pass across multi-company and Joint Venture scenarios.
- Dashboards, reports, exports and mobile layouts pass accessibility and responsive QA.
- Performance and failure-isolation targets are met under representative load.
- Data quality, lineage, refresh status, audit and observability are production ready.
- Deployment, rollback, rebuild, backfill and disaster-recovery runbooks are approved.
- No unresolved critical security, privacy or data-reconciliation defects remain.

Final product position: The Reporting & Analytics Module is the trusted analytical layer that converts every ConstructionOS operational record into governed, traceable and permission-safe business intelligence.